

Economic-Aware and Sovereignty-Constrained Routing for Enterprise LLM Gateways

Draft v0.4 — two-provider (OpenAI US + Mistral EU) evaluation with a 30-repetition statistical run.

A complete
reproduction

Abstract

Enterprises are adopting large language models (LLMs) faster than they can govern them: they cannot

say which
model,

1. Introduction

LLMs are now embedded in HR assistants, document RAG, developer tooling, and analytical agents. Each

call has a

1. **Cost opacity.** Teams cannot attribute spend per application/team or decide when a cheaper model

would suff

Existing gateways and FinOps tools mostly *measure and report*. Our thesis is that a control plane at

the **gatew**

Contributions.

- A cons

2. Background and Related Work

References are in `references.bib` (recent, verified); citation keys are shown in brackets.

Model routing and cascades. A growing line of work trades quality for cost by selecting among

models pe
a

LLM serving systems. Orca [yu2022orca], vLLM/PagedAttention [kwon2023pagedattention], SGLang

[zheng2023]

LLM gateways. Envoy AI Gateway [envoyaigateway] and LiteLLM [litellm] provide a unified data path

to multiple

Evaluation. We follow the LLM-as-a-judge methodology of Zheng et al. [zheng2023judge] (MT-Bench /

Chatbot A we adopt

Regulation. Data-residency and sensitive-data handling are driven by the GDPR [gdpr2016] and the

EU AI Act [euaiact2024]

Positioning (Table 0). The following feature matrix makes the delta explicit: prior routing work

optimizes

Capability	FrugalGPT	Hybrid LLM	RouteLLM	AutoMix	**Ours**
Cost-aware routing	✓	✓	✓	✓	✓
Quality-aware routing	✓	✓	✓	✓	✓
Learned difficulty predictor	✓	✓	✓	✓	✗ (priors; future: graft)
Hard sovereignty constraints	✗	✗	✗	✗	**✓**
Per-team/namespace budgets	✗	✗	✗	✗	**✓**
Budget-aware graceful degradation	✗	✗	✗	✗	**✓**
Cost attribution (team/ns/app)	✗	✗	✗	✗	**✓**
Managed-vs-self-hosted break-even	✗	✗	✗	✗	**✓**
Gateway/K8s control-plane artifact	✗	✗	✗	✗	**✓**

We do not claim to beat learned routers on the cost-quality frontier; rather, our governance layer is

*orthogonal

Side-by-side comparison (Table 0b). Against the systems most often cited in this space — three

learned routers

Dimension	RouteLLM	FrugalGPT	AutoMix	LiteLLM	**Ours**
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Type	learned router	cascade	self-verify router	gateway/proxy	**governance control plane**
Primary goal	cost↔quality	cost↔quality	cost↔quality	unified access + basic cost log	**cost under governance**
Routing signal	learned (preference data)	confidence cascade	few-shot self-verification + POMDP	static config / simple fallback	priors + policy (learned = future)
Multi-provider	partial	yes	yes	**yes**	**yes (OpenAI + Mistral EU)**
Hard sovereignty constraints	x	x	x	x	**✓ (zones, sensitive data)**
Per-tenant budgets + degradation	x	x	x	basic budgets, no degradation	**✓ (graceful)**
Cost attribution (team/ns/app)	x	x	x	per-key spend	**✓ (policy-level)**
Managed-vs-self-hosted break-even	x	x	x	x	**✓ (modeled)**
Form factor	library	library	library	proxy (data path)	**K8s operator + Envoy data path**

LiteLLM is the closest *engineering* peer (a real gateway with per-key spend and fallbacks) but offers no constraints

3. Problem Formulation and Algorithm

A request r carries metadata (team, namespace, sensitivity) and an estimated token footprint. Given a catalog

Hard constraint. Reject any m that violates P : its zone is forbidden; or an allow-list is set and m 's

Soft objective. Among C meeting a minimum quality $q_{\min}$ (relaxed only when the budget is exhausted

$$\arg\min_{m \in C} \alpha \cdot \widehat{c}(m) \cdot (1 + \epsilon \cdot \rho) + \beta \cdot \ell_q(m) + \gamma \cdot \ell_\lambda(m) + \delta \cdot \sigma(m)$$

where $\widehat{c}(m) = \frac{\text{cost}(m)}{\text{cost}(\text{premium})}$ is normalized cost, $\rho = \text{used}/\text{total}$ is budget expected

Pseudocode and the exact normalization are in `methodology.md`; the implementation is `experiment`

4. System Design

Requests flow client → **Envoy AI Gateway** → provider (managed API or self-hosted). The **AI**

5. Methodology

Testbed. A Go harness drives the operator's **actual** engines and issues **real** LLM calls across

two prov
gpt-4.1-na

Workloads (W1-W4). HR chatbot (short, sensitive), RAG docs (long context, quality-sensitive), Dev

assistant (

Baselines. B1 premium-static, B2 round-robin, B3 least-cost, B4 static namespace policy, B5 budget

hard-block

Metrics. Cost (total, per request, per token, per team via the operator cost engine, % savings);

quality (LL
the

Reproducibility. Temperature 0; all responses and judge scores cached (`results/cache.json`) so

re-runs are

6. Results

RQ1 — Cost (Figure 2, Table 1)

Relative to

RQ2 — Quality (Figure 3, Table 2)

Within this

RQ2b — Objective accuracy on public benchmarks + judge validity

To remove
250

Why this contradicts the judge (a methodological finding). We re-scored 300 objective answers with

both ex
81.1% of

RQ3 — Latency and overhead (Figure 4, Table 3)

The routing

Gateway data-path overhead (real Envoy, under load). Beyond control-plane decision time, we

measured

RQ4 — Sovereignty (Figure 5, Table 4)

A sovereign

RQ5 — Budget-aware degradation (Figure 6, Table 5)

Under a b
availabili

RQ6 — Managed vs self-hosted break-even (Figure 7) — *modeled*

Using rea
managed

Ablation (Figure 8, Table 7)

Removing
term

Statistical significance (N=30 repetitions)

To move b

Figures `figures/fig\_stats\_{cost,quality,latency}.png` show means $\pm 95\%$ CI over the 30 repetitions.

7. Discussion

Economic-aware routing is most valuable when a workload mixes easy and hard requests: the router sends

cheap req

When **not** to use economic routing

Economic

8. Threats to Validity

Summarized here; full version in `threats\_to\_validity.md`. **Internal:** API/latency variability

(mitigated

9. Conclusion and Future Work

Within our preliminary, two-provider, single-pass testbed, a gateway-level, economic-aware,

sovereignty
keeping

Artifact / Reproducibility

Operator + `experimentation/` (datasets, harness, scripts, results, figures, cached responses).

Reproducibility

References

See `references.bib` (BibTeX). Key works cited: model routing/cascades — FrugalGPT

[chen2023]

Appendix: Figures

Figure 2 — Total cost by routing strategy

0.040
0.035
0.030
0.025
0.020
0.015
0.010
0.005
0.000

Total cost (EUR)

Figure 3 — Quality vs cost

0.91
0.90
0.89
0.88
0.87
0.86

Mean quality (normalized 0-1)

Figure 4 — Latency p95 and mean $\pm$ 95% CI

3000 -
2500 -
2000 -
1500 -
1000 -
500 -
0 -

Latency (ms)

Figure 5 — Sovereignty: cost & violations

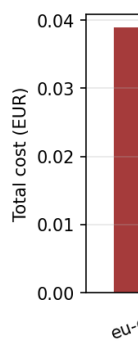


Figure 6 — Budget policy: availability vs overrun



Figure 7 — Managed vs self-hosted break-even (modeled)

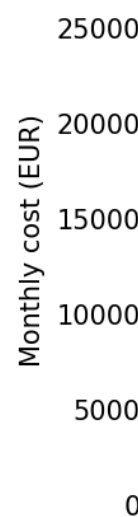


Figure 8 — Ablation of scoring terms

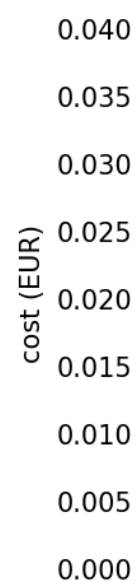


Figure S1 — Cost per rep, mean $\pm$ 95% CI (N=30)

0.040
0.035
0.030
0.025
0.020
0.015
0.010
0.005
0.000

EUR

Figure S2 — Quality per rep, mean $\pm$ 95% CI (N=30)

1.0
0.8
0.6
0.4
0.2
0.0

quality (norm)

Figure S3 — Latency per rep, mean $\pm$ 95% CI (N=30)

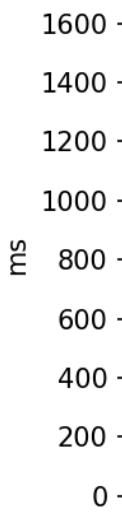


Figure S4 — LLM-judge score by objective correctness (judge over-rates wrong answers)

